Huawei Access Node Optical Transport Failure Resolution SOP (BAS-ACC-53, Basavanagudi POP)

Node: BAS-ACC-53 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_8040

Description

This Standard Operating Procedure outlines the steps to diagnose and resolve optical transport failures in Huawei Access Nodes, specifically in the context of Node ID BAS-ACC-53 situated at the Basavanagudi POP. The failure domain under investigation encompasses SFP/XFP degradation, fiber cuts, and Rx/Tx power loss. The Access Node's Optical Transport Network (OTN) plays a critical role in transporting client signals over the optical link. The OTN layer is responsible for multiplexing and demultiplexing client signals onto a single wavelength or multiple wavelengths, respectively. In the event of an optical transport failure, it is essential to identify the root cause of the issue, which could be attributed to a faulty SFP/XFP, a fiber cut, or a power loss in the receive or transmit direction. Huawei's CLI dialect is employed throughout this SOP for all diagnostic and resolution commands.

Pre-requisite Checks

- check: Verify the redundant link status to ensure continued network connectivity in case of the primary link failure, command: display link-info

- check: Dispatch remote hands to the Basavanagudi POP for physical access to the node, command: None

Potential Root Causes

Cause	Probability
Faulty SFP/XFP	High
Fiber cut	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display sfp info, Expected Output: SFP information will be displayed, including the SFP's status, type, and optical characteristics, Description: This command isolates the SFP's operational status and optical characteristics, helping to identify potential issues with the SFP itself or the optical link.

Command: display fiber info, Expected Output: Fiber information will be displayed, including the fiber's status, type, and attenuation characteristics, Description: This command isolates the fiber's operational status and attenuation characteristics, helping to identify potential issues with the fiber itself or the optical link.

Command: display rx-tx power info, Expected Output: Rx/Tx power information will be displayed, including the receive and transmit
```

```
power levels, Description: This command isolates the receive and
transmit power levels, helping to identify potential issues with the
node's optical power budget or the optical link.
```

Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP, Command: replace sfp, Description: Physically replace the faulty SFP/XFP with a spare unit, ensuring that the new SFP/XFP is properly seated and secured.

Step 2: Step: Repair or replace the fiber, Command: display fiber info, Description: If the fiber is damaged or cut, repair or replace the fiber to restore optical connectivity.

Step 3: Step: Adjust the node's optical power budget, Command: set rx-tx power, Description: Adjust the node's optical power budget to ensure that the receive and transmit power levels are within the recommended ranges.

Post-Check Verification

```
Command: display sfp info, Expected Output: The replaced SFP/XFP
should be displayed as operational, with its status set to 'up',
Description: Verify that the replaced SFP/XFP is operational and its
status is set to 'up'.

Command: display fiber info, Expected Output: The repaired or replaced
fiber should be displayed as operational, with its status set to 'up',
Description: Verify that the repaired or replaced fiber is operational
and its status is set to 'up'.

Command: display rx-tx power info, Expected Output: The receive and
transmit power levels should be within the recommended ranges,
Description: Verify that the receive and transmit power levels are
within the recommended ranges, indicating that the node's optical
power budget is properly adjusted.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: replace sfp, Description: Physically replace the new SFP/XFP with the original faulty unit, ensuring that the original SFP/XFP is properly seated and secured.

- Step: Restore the original fiber, Command: display fiber info, Description: If the fiber was repaired or replaced, restore the original fiber to its previous state.

- Step: Restore the original node configuration, Command: undo set rx-tx power, Description: Restore the original node configuration, including the optical power budget, to its previous state.